Graph Theory

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HKIMO Training Level 2

1 Definitions

- A **graph** G consists of a set of **vertices** V and a set of **edges** E joining pairs of vertices.
- A **loop** is an edge where both endpoints are the same.
- A graph is **simple** if it has no loops, and there is no more than one edge connecting each pair of distinct vertices.
- Two vertices $v, w \in V$ are **adjacent** if there is an edge connecting v and w.
- Two edges are **incident** if they are both connected to the same vertex.
- The **degree** of a vertex v, denoted by $d(v)$, is the number of edges which contain v as an endpoint.
- A **path** in G is a finite sequence of distinct vertices, $v_1, \cdots, v_k$ such that v_i is adjacent to v_{i+1}. The **length** of a path is the number of edges in the path.
- The **distance** between two vertices is the minimum possible length of the paths connecting the vertices.
- A **cycle** is a path where the first and last vertices are also adjacent.
- A graph is **connected** if for every pair of distinct vertices v, w there exists a path from v to w. If a graph is not connected, it is **disconnected**.
- A graph G is a **tree** if it is connected and contains no cycles. If a tree has n vertices, then it has $n-1$ edges.
- A graph of n vertices is **complete** and denoted by K_n if there is an edge between all $\binom{n}{2}$ pairs of vertices.
- A graph is **bipartite** if the vertex set V can be partitioned into two sets A and B such that there is no edge that has both endpoints in the same set.
- If $V' \subset V$, the **induced subgraph** of V' is the graph containing the vertices V' and all the edges with both endpoints in V'.
- A **clique** is a subset of the vertices such that the induced subgraph of the subset is complete.
- A **maximal clique** is a clique that is not a subgraph of a larger clique.
- A **connected component** is a *maximal* set of vertices in G such that the induced subgraph of G is connected. A graph can be partitioned into connected components.

2 Selected Problems

Problem 1 (HKTST 2017). In a committee there are n members. Each pair of members are either friends or enemies. Each committee member has exactly three enemies. It is also known that for each committee member, an enemy of his friend is automatically his own enemy. Find all possible value(s) of n.

Problem 2 (HKTST 2018). In a group of 2017 persons, any pair of persons has exactly one common friend (other than the pair of persons). Determine the smallest possible value of the difference between the numbers of friends of the person with the most friends and the person with the least friends in such a group.

Problem 3 (CHKMO 2017). A, B and C are three persons among a set P of n ($n > 3$) persons. It is known that A, B and C are friends of one another, and that every one of the three persons has already made friends with more than half the total number of people in P. Given that every three persons who are friends of one another form a friendly group, what is the minimum number of friendly groups that may exist in P?

Problem 4 (Canada 2006). Consider a round-robin tournament with $2n+1$ teams, where each team plays each other team exactly one. We say that three teams X, Y and Z, form a cycle triplet if X beats Y, Y beats Z and Z beats X. There are no ties.

1. Determine the minimum number of cycle triplets possible.
2. Determine the maximum number of cycle triplets possible.

Problem 5 (APMO 2016). The country Dreamland consists of 2016 cities. The airline Starways wants to establish some one-way flights between pairs of cities in such a way that each city has exactly one flight out of it. Find the smallest positive integer k such that no matter how Starways establishes its flights, the cities can always be partitioned into k groups so that from any city it is not possible to reach another city in the same group by using at most 28 flights.

Problem 6 (IMO 1991). Suppose G is a connected graph with k edges. Prove that it is possible to label the edges $1, 2, \ldots, k$ in such a way that at each vertex which belongs to two or more edges, the greatest common divisor of the integers labeling those edges is equal to 1.

Problem 7 (IMO 1992). Consider 9 points in space, no four of which are coplanar. Each pair of points is joined by an edge (that is, a line segment) and each edge is either colored blue or red or left uncolored. Find the smallest value of n such that whenever exactly n edges are colored, the set of colored edges necessarily contains a triangle all of whose edges have the same color.

Problem 8 (IMO Shortlist 2002). Among a group of 120 people, some pairs are friends. A weak quartet is a set of four people containing exactly one pair of friends. What is the maximum possible number of weak quartets?

Problem 9 (IMO 2007). In a mathematical competition some competitors are friends. Friendship is always mutual. Call a group of competitors a clique if each two of them are friends. (In particular, any group of fewer than two competitors is a clique.) The number of members of a clique is called its size.

Given that, in this competition, the largest size of a clique is even, prove that the competitors can be arranged into two rooms such that the largest size of a clique contained in one room is the same as the largest size of a clique contained in the other room.

Problem 10 (IMO 2019). A social network has 2019 users, some pairs of whom are friends. Whenever user A is friends with user B, user B is also friends with user A. Events of the following kind may happen repeatedly, one at a time:

- Three users A, B, and C such that A is friends with both B and C, but B and C are not friends, change their friendship statuses such that B and C are now friends, but A is no longer friends with B, and no longer friends with C. All other friendship statuses are unchanged.

Initially, 1010 users have 1009 friends each, and 1009 users have 1010 friends each. Prove that there exists a sequence of such events after which each user is friends with at most one other user.